Executive Summary

This analysis evaluates 817,860.05 in restaurant revenue across 21,350 transactions using advanced SQL techniques (CTEs, window functions, RANK, LAG, NTILE).

Key findings include:

- Strong demand concentration during lunch (12–1 PM) and dinner (5–7 PM) hours.
- Large pizzas generate 45.89% of total revenue, making size the primary revenue driver.
- Revenue distribution across categories is balanced (23–27%), reducing dependency risk.
- Top 20% of pizzas contribute approximately 33% of revenue, indicating diversified product performance.
- Chicken-based pizzas consistently dominate monthly rankings.

Strategic recommendations include peak-hour staffing optimization, pricing strategy adjustments for Large sizes, and targeted promotions during weaker seasonal months.

1. Project Title

Restaurant Sales & Demand Analysis Using Advanced SQL

2. Project Objective

The objective of this project is to conduct a structured sales and demand analysis using MySQL to identify revenue drivers, seasonal fluctuations, product performance trends, and operational optimization opportunities across $817K+ in annual restaurant revenue.

3. Dataset Description

The dataset consists of four relational tables:

- orders – Contains order date and time
- order_details – Contains pizza items within each order
- pizzas – Contains pizza size and price
- pizza_types – Contains pizza name, category, and ingredients

The tables are connected using primary and foreign keys, forming a structured relational schema suitable for transactional sales analysis.

4. Phase 1 – Revenue Foundation Analysis

4.1 Total Revenue

The total revenue generated during the recorded period is **817,860.05**.
This serves as the baseline performance indicator for evaluating sales growth and product contribution.

total_revenue
817860.05

4.2 Total Orders

A total of **21,350 orders** were recorded during the analysis period.

This metric reflects overall transaction volume and helps evaluate customer purchasing frequency.

total_orders
21350

4.3 Average Order Value (AOV)

The average order value is **38.31**.

This indicates that, on average, each customer transaction generates approximately 38 in revenue. This suggests that customers typically purchase multiple items per order rather than a single low-cost product.

avg_order_value
38.31

4.4 Average Pizzas per Order

On average, customers purchase **2.32 pizzas per order**.

This confirms that most transactions involve multiple items, contributing to the average order value of 38.31. The data suggests strong bundled purchasing behavior, which may indicate group orders or family-sized consumption patterns.

avg_pizzas_per_order
2.32

5. Phase 2 – Time-Based Analysis

5.1 Monthly Revenue Trend

Revenue was aggregated by month to evaluate seasonal performance patterns.

Month	Revenue ($)
Jan	69,793.30
Feb	65,159.60
Mar	70,397.10
Apr	68,736.80
May	71,402.75
Jun	68,230.20
Jul	72,557.90
Aug	68,278.25
Sep	64,180.05
Oct	64,027.60
Nov	70,395.35
Dec	64,701.15

month	monthly_revenue
2015-01	69793.3
2015-02	65159.6
2015-03	70397.1
2015-04	68736.8
2015-05	71402.75
2015-06	68230.2
2015-07	72557.9
2015-08	68278.25
2015-09	64180.05
2015-10	64027.6
2015-11	70395.35
2015-12	64701.15

Key Observations:

- Revenue peaks in **July**, suggesting strong mid-year demand.
- Revenue declines in **September and October**, indicating possible seasonal slowdown.
- Sales recover in **November**, potentially due to holiday or promotional activity.

5.2 Month-over-Month Revenue Growth

Revenue growth was calculated using the LAG() window function to measure month-over-month performance changes.

month	revenue	previous_month	revenue_change
2015-01	69793.3	NULL	NULL
2015-02	65159.6	69793.3	-4633.7
2015-03	70397.1	65159.6	5237.5
2015-04	68736.8	70397.1	-1660.3
2015-05	71402.75	68736.8	2665.95
2015-06	68230.2	71402.75	-3172.55
2015-07	72557.9	68230.2	4327.7
2015-08	68278.25	72557.9	-4279.65
2015-09	64180.05	68278.25	-4098.2
2015-10	64027.6	64180.05	-152.45
2015-11	70395.35	64027.6	6367.75
2015-12	64701.15	70395.35	-5694.2

Key Findings:

- The highest revenue increase occurred in **November (+6,367.75)**, indicating a strong late-year demand surge.
- Revenue peaked in **July**, suggesting mid-year seasonality.
- Significant revenue declines were observed in **December (-5,694.20)** and **February (-4,633.70)**.
- Revenue trends show fluctuations rather than consistent growth, indicating seasonal or event-driven demand patterns.

These findings highlight the importance of strategic promotions during weaker months and inventory planning during peak periods.

5.3 Revenue by Day of Week

Revenue was analyzed by day of the week to evaluate weekly demand patterns.

Day	Revenue ($)
Friday	136,073.90
Thursday	123,528.50
Saturday	123,182.40
Wednesday	114,408.40
Tuesday	114,133.80
Monday	107,329.55
Sunday	99,203.50

day_name	revenue
Friday	136073.9
Thursday	123528.5
Saturday	123182.4
Wednesday	114408.4
Tuesday	114133.8
Monday	107329.55
Sunday	99203.5

Key Observations:

- **Friday generates the highest revenue**, indicating strong end-of-week demand.
- Contrary to common assumptions, **Sunday generates the lowest revenue**.
- Revenue gradually increases from Monday through Friday, suggesting weekday momentum buildup.

These insights can support staffing optimization and targeted weekday promotions.

5.4 Peak Demand Hours

Order volume was analyzed by hour of the day to identify peak demand periods.

Hour	Total Orders
12 PM	2,520
1 PM	2,455
6 PM	2,399
5 PM	2,336
7 PM	2,009

hour_of_day	total_orders
12	2520
13	2455
18	2399
17	2336
19	2009

Key Observations:

- The highest order volume occurs during **12 PM–1 PM**, indicating a strong lunch rush.
- A second peak occurs between **5 PM–7 PM**, reflecting dinner demand.
- Orders decline significantly after 9 PM, with minimal late-night activity.

These findings suggest the need for optimized staffing during lunch and dinner hours, while late-night operations may not require full capacity.

6. Phase 3 – Product Performance Analysis

6.1 Top 10 Pizzas by Revenue

Revenue contribution was analyzed at the product level to identify top-performing pizzas.

Pizza	Revenue ($)
Thai Chicken	43,434.25
Barbecue Chicken	42,768.00
California Chicken	41,409.50
Classic Deluxe	38,180.50
Spicy Italian	34,831.25
Southwest Chicken	34,705.75
Italian Supreme	33,476.75
Hawaiian	32,273.25
Four Cheese	32,265.70
Sicilian	30,940.50

name	revenue
The Thai Chicken Pizza	43434.25
The Spicy Italian Pizza	34831.25
The Southwest Chicken Pizza	34705.75
The Sicilian Pizza	30940.5
The Italian Supreme Pizza	33476.75
The Hawaiian Pizza	32273.25
The Four Cheese Pizza	32265.7
The Classic Deluxe Pizza	38180.5
The California Chicken Pizza	41409.5
The Barbecue Chicken Pizza	42768

Key Observations:

- Chicken-based pizzas dominate the top revenue rankings.
- The Thai Chicken Pizza generates the highest revenue overall.
- Revenue appears concentrated among a small group of high-performing products.

6.2 Pareto Analysis – Revenue Concentration

To evaluate revenue concentration, pizzas were divided into five equal groups (top 20% each) using the NTILE() window function.

name	revenue	revenue_percentage	revenue_group
The Thai Chicken Pizza	43434.25	5.31	1
The Barbecue Chicken Pizza	42768	5.23	1
The California Chicken Pizza	41409.5	5.06	1
The Classic Deluxe Pizza	38180.5	4.67	1
The Spicy Italian Pizza	34831.25	4.26	1
The Southwest Chicken Pizza	34705.75	4.24	1
The Italian Supreme Pizza	33476.75	4.09	1

The top 20% of pizzas contribute approximately 33% of total revenue.

Key Insights:

- Revenue is moderately concentrated among high-performing products.
- The business is not overly dependent on a single product.
- However, top-performing pizzas should remain marketing priorities.
- Diversification across products appears relatively balanced.

This indicates a healthy revenue distribution, reducing dependency risk while still highlighting core revenue drivers.

6.3 Revenue by Category

Revenue was analyzed across product categories to evaluate category-level contribution.

Category	Revenue ($)	Revenue %
Classic	220,053.10	26.91%
Supreme	208,197.00	25.46%
Chicken	195,919.50	23.96%
Veggie	193,690.45	23.68%

category	revenue	revenue_percentage
Classic	220053.1	26.91
Supreme	208197	25.46
Chicken	195919.5	23.96
Veggie	193690.45	23.68

Key Insights:

- Revenue distribution across categories is highly balanced.
- The Classic category generates the highest share (26.91%), but the difference across categories is minimal.
- No single category dominates overall sales.
- This indicates a diversified product portfolio with reduced dependency risk.

6.4 Revenue by Size

Revenue contribution was analyzed by pizza size to evaluate customer purchasing preferences.

Size	Revenue ($)	Revenue %
L	375,318.70	45.89%
M	249,382.25	30.49%
S	178,076.50	21.77%
XL	14,076.00	1.72%
XXL	1,006.60	0.12%

size	revenue	revenue_percentage
L	375318.7	45.89
M	249382.25	30.49
S	178076.5	21.77
XL	14076	1.72
XXL	1006.6	0.12

Key Insights:

- Large pizzas generate nearly **46% of total revenue**, making them the primary revenue driver.
- Medium pizzas contribute approximately 30%, reinforcing strong demand for mid-to-large portion sizes.
- XL and XXL sizes contribute negligibly to total revenue.
- Revenue is highly concentrated in standard sizes (S, M, L).

6.5 Best-Selling Pizza by Month

To identify consistent top performers, monthly revenue rankings were calculated using the RANK() window function.

month	name	revenue	rank_position
2015-01	The Barbecue Chicken Pizza	3770.25	1
2015-02	The California Chicken Pizza	3537	1
2015-03	The Barbecue Chicken Pizza	4049.25	1
2015-04	The Barbecue Chicken Pizza	3788.5	1
2015-05	The Barbecue Chicken Pizza	3953.5	1
2015-06	The California Chicken Pizza	3870.5	1
2015-07	The Thai Chicken Pizza	4073.75	1
2015-08	The California Chicken Pizza	3977.25	1
2015-09	The Thai Chicken Pizza	3736.25	1
2015-10	The Thai Chicken Pizza	3503.25	1
2015-11	The Barbecue Chicken Pizza	3881.75	1
2015-12	The Thai Chicken Pizza	3813	1

Findings:

- Chicken-based pizzas (Barbecue Chicken, Thai Chicken, and California Chicken) consistently ranked as the top-selling pizzas across most months.
- No veggie or classic pizza dominated on a monthly basis.
- This indicates strong and stable consumer preference for chicken-based offerings.

Strategic Implication:

- Chicken pizzas should remain central to marketing campaigns.
- Promotional bundles could be built around top-performing chicken variants.
- Inventory planning should prioritize high-demand chicken ingredients.

Final Section: Business Recommendations

Strategic Recommendations

1. Optimize staffing between 12–1 PM and 5–7 PM to match peak operational demand.
2. Consider targeted promotions during September and October to mitigate seasonal slowdowns.
3. Evaluate price elasticity for Large pizzas, given their 45.89% revenue contribution.
4. Prioritize chicken-based variants in marketing campaigns due to consistent monthly dominance.
5. Reassess operational necessity of XL and XXL sizes due to minimal revenue contribution.
6. Maintain product portfolio diversity to preserve revenue stability.